

DEPARTMENT OF SMART COMPUTING AND CYBER RESILIENCE

FACULTY OF ENGINEERING AND TECHNOLOGY

FINAL PROJECT FOR CSC1024\PRG3024\FEL1184 PROGRAMMING PRINCIPLES

ACADEMIC SESSION: May 2026

DISTRIBUTION: WEEK 06

SUBMISSION DEADLINE: 2nd August 2026 (Sunday), before 11:59 PM.

Student Names with IDs:

1. Amos Lo Jun Hao (25018961)

(Group Leader)

2. Law Yuan Rui (26035766)

3. Wong Xin Er (23088320)

4. Arooba Shahid (26011023)

INSTRUCTIONS TO CANDIDATES

- This programming project will contribute 50% to your final grade.
- This programming project is a final examination.
- This is a group project with a maximum of 5 members.
- The presentation will be carried out in Week 13 during lab slots (Please email your lab instructor to book your presentation slot)
- The group leader will submit the following materials via elearn:
 - ✓ Report in .pdf
 - ✓ Contribution log signed by each member of the group
 - ✓ Program source file in either .py or .ipynb, including .txt files
 - ✓ Link to the recorded presentation. Make sure the accessibility is set to "Everyone".
 - ✓ Presentation deck in .pdf
- Please ensure that for both report and presentation deck submitted, each student name and ID are included. Also, do note that marks for the presentation may differ from each member based on their understanding and contribution to the project.

IMPORTANT

The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

- Project submitted after the deadline but within 1 week will be accepted for a maximum mark of 40%.
- Work handed in following the extension of 1 week after the original deadline will be regarded as a non-submission and marked **ZERO**.

Lecturer's Remark (Use additional sheet if required)

We, the students as presented above, have received the programming project and read the comments.

Four handwritten signatures in black ink, each followed by a dotted line for a date.

(Student signatures and Date)

Academic Honesty Acknowledgement

We, the students as presented above, verify that this paper contains entirely our own work. We have not consulted with any outside person or materials other than what was specified (such as an interviewee) in the assignment or the syllabus requirements. Further, we have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. We understand the penalties (as outlined in the student handbook) for any form of copying or collaboration on any assignment.

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(Student signatures and Date)

CSC1024 Programming Principles

Final Project



Python-Powered Social Media Content Planner



Project Description:

Social media is central to Gen Z life, from personal branding to content creation. This project challenges you to build a Social Media Content Planner, a text-based application that helps content creators plan, track, and analyse their posts across multiple platforms. The system will manage post ideas, schedule content, track engagement metrics, and generate performance summaries, all powered by Python and plain text files.



Project Requirements:

- a. Initial Data Preparation:** Create the following text files to serve as the system's persistent data storage:
 - ✓ [posts.txt](#) – Fields: Post ID, platform (e.g. Instagram, TikTok, X), content caption, scheduled date, status (Draft/Scheduled/Posted).
 - ✓ [platforms.txt](#) – Fields: Platform ID, platform name, follower count.
 - ✓ [engagement.txt](#) – Fields: Post ID, likes, comments, shares, views.
- b. Program Functionality:** Implement a user-friendly menu system with the following options (at least):
 - ✓ [Add a new post idea](#): Allow the user to draft a new post by entering the Post ID, target platform, content caption, and scheduled date. Status is set to Draft by default.
 - ✓ [Update post status](#): Enable the user to change the status of a post from Draft to Scheduled, or from Scheduled to Posted.
 - ✓ [Record engagement metrics](#): Allow the user to log engagement data (likes, comments, shares, views) for a posted piece of content.
 - ✓ [Display content calendar](#): Show all posts sorted by scheduled date, displaying the platform, caption preview, and current status.
 - ✓ [Generate performance report](#): Display a summary including total posts per platform, the best-performing post (highest engagement), and the platform with the most interaction.

- ✓ [*Export report to file:*](#) Save the performance summary to a new text file (e.g. report.txt) for sharing or record keeping.
- ✓ [*Exit:*](#) Provide an option to exit the program gracefully.

c. Programming Techniques: Your solution must demonstrate the following programming techniques:

- ✓ Use Command Line Interfacing.
- ✓ Use input and display functions for user interaction and data input/output.
- ✓ Employ file handling for data persistence and retrieval.
- ✓ Use conditional statements and loops for decision-making and repeated tasks.
- ✓ Define functions for modularity and reusability.

d. Input and Output Handling Requirements: Your program must handle invalid or unexpected user input without crashing and must produce clear, correct, and meaningful output. Every menu option, data entry point, and displayed result should include appropriate validation, error handling, and formatting. Students are expected to think critically about possible input errors, edge cases, incorrect outputs, and misuse scenarios, and design solutions to prevent failures and ensure smooth program execution.



Marking Rubrics (Total Marks: 100):

Group Evaluation (70%)

Assessment Criterion	Marks
Code Organisation and Readability (5 marks)	
Proper indentation and consistent coding style	2
Meaningful variable and function names	1
Appropriate use of comments to enhance code understanding	1
Logical and efficient code structure	1
Functionality and Correctness (25 marks)	
Accurate implementation of all main functionalities	5
Correct data validation and logical operations	5
Proper handling of updates and deletions	10
Display of information with accuracy and clarity	5
Data Persistence and File Handling (10 marks)	
Proper creation and utilisation of text files for data storage	5
Accurate reading and writing of data to and from files	5
User Interface and User Experience (8 marks)	
User-friendly menu system	2
Clear and concise prompts and messages	2
Smooth navigation and flow of the program	4
Error Handling and Robustness (8 marks)	
Appropriate error handling for input validation	4
Graceful handling of edge cases and exceptions	2

Assessment Criterion	Marks
Overall program stability and reliability	2
Flowchart (4 marks)	
Clarity and readability of the flowchart	1
Accurate representation of the program flow	1
Inclusion of all major processes and decision points	1
Proper use of flowchart symbols and conventions	1
Report (10 marks)	
Clarity and organisation of the report	2
Detailed explanation of the system design and architecture	2
Discussion of the implementation challenges and solutions	2
Analysis of the system's strengths and limitations	2
Suggestions for future improvements and enhancements	2

Individual Evaluation (30%)

Assessment Criterion	Marks
Presentation (10 marks)	
Clarity and effectiveness of the presentation	2
Demonstration of the system's functionality	3
Explanation of the key features and design decisions	3
Time management and adherence to the allotted presentation duration	2
Contribution (20 marks)	
Code Contribution: Student's actual code ownership, logic accuracy, and integration	5

Assessment Criterion	Marks
Understanding of Code: Ability to explain and justify written code during viva/reflection	10
Team Collaboration: Participation, communication, and contribution logs	4
Professionalism & Responsibility: Commitment, documentation effort, and ethics	1